

CASSANDRA RISK — MASTER CONTEXT DOCUMENT v2.0

Thread Date: 2026-04-02 | Mumbai IST
Status: Living document — update after every major session
Repo: github.com/umran-n/cassandra-risk-replication
Version: v2.0 (supersedes CASSANDRA RISK — FULL SESSION CONTEXT v0.5.8)

1. PROJECT IDENTITY

Project Name: Cassandra Risk
Tagline: Bloomberg tells you what happened. Cassandra tells you what's coming.
Core Hypothesis: Prediction market event probabilities contain forward-looking regime fragility signals that can govern portfolio exposure more efficiently than backward-looking risk controls (VaR, realised volatility).
Framework: Regime Stability Index (RSI) computed from weighted, horizon-decayed event probabilities across hazard categories. Portfolio position = RSI. When aggregate event hazard rises, exposure falls.
Epistemological Arbitrage: At the intersection of prediction markets (The Gambler), forecasters (The Philosopher), institutional quants (The Bureaucrat), and academics (The Validator) — none of whom talk to each other.

2. PUBLISHED RESEARCH (DOI-STAMPED)

Paper	Title	DOI	Status
Paper 1	Beyond Value-at-Risk: Cassandra-Risk via Prediction Markets	10.13140/RG.2.2.21272.05124	✔ Live
Paper 2	Governance & Calibration Architecture (Becker Stack)	10.13140/RG.2.2.17209.12644	✔ Live
Paper 3	Ensemble Signal: Polymarket + Kalshi Cross-Platform Validation	TBD	🔨 In progress
Paper 4	Fragility Alpha: RSI vs VaR Delta (Bigdata.com cross-validation)	TBD	📋 Planned

Paper 5	Kelly-Optimal Position Sizing from Governed RSI	TBD	 Planned
Paper 6	Asset Class RSI Vector & Event-Driven Rotation	TBD	 Planned
Paper 7	On-Chain Funds Movement as Cross-Signal Alpha	TBD	 Planned
Paper 8	SIR Contagion Model with Prediction Market Transmission Coefficient	TBD	 Planned

Rule: Every API tier has a corresponding paper. Every paper funds the next API tier.

3. HEADLINE PERFORMANCE (Paper 1 Baseline — V4)

Metric	Buy & Hold	Vol Targeting	Cassandra V4
CAGR	13.99%	11.46%	15.99%
Sortino	0.733	0.836	1.159
Daily MDD	-33.72%	-15.14%	-20.27%
Avg Position	100%	77.3%	78.7%
Total Return (5yr)	92.73%	72.22%	110.31%

Backtest period: 2020-01-01 to 2025-01-10, SPY daily, 3 independent replication passes.
Key result: Higher return AND lower drawdown simultaneously — not a conventional risk/return tradeoff. A new information dimension.

4. CURRENT BEST ARCHITECTURE (Frozen at v0.5.7)

```

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Strategy:      V5-Becker-top5cap-geo
Tag:           v0.5.7-geo-subbucket-calibration
Commit:       8ec3f50
Performance:   Sortino 0.330 | CAGR 7.24% | MDD -33.72%

Architecture layers:
  Monetary:    Becker 0.0017 + top-5 removal + 30% cap
  Geopolitical: Governed admission (500K floor, binary,
macro-relevant, 25% cap), UNCALIBRATED

```

All other: Standard admission, no calibration correction yet

Becker Efficiency Gap Constants

Category	Gap	Source
monetary_policy	0.0017	Kalshi, 72.1M trades, \$18.26B volume
geopolitical	0.0732	Widest — heterogeneous event types
electoral	0.0102	—
trade_technology	0.0269	—
fiscal_debt	0.0102	—
systemic_credit	0.0102	—

5. BUILD VERSION HISTORY (Key Milestones)

Version	Description	Sortino	CAGR
v0.5.0	V5 Expansion — 38-event Polymarket universe	0.231	5.75%
v0.5.3	Becker calibration layer added	0.241	5.89%
v0.5.4	Becker + top-5 + cap30 stack	0.323	7.13%
v0.5.6	Geopolitical expansion (uncalibrated)	0.330	~7.2%
v0.5.7	Sub-bucket geo calibration (frozen best)	0.330	7.24%
v0.5.8	Monte Carlo bootstrap CI (in progress at session end)	—	—
v0.6.x	Kalshi dredger + governed promotion workflow	 Built	—
v1.0.0	Live RSI API — deployed to Railway	 LIVE	—

6. LIVE API STATUS

Marketplace	Status	Price
RapidAPI	 Live	\$49/mo PRO
Zyla Labs	 Live	Active

Postman  —
Deferred

AWS Marketplace  Planned Month 2

Azure Marketplace  Planned Month 2

Subscribers: 1 (RapidAPI)

MCP Access: Zyla auto-exposes MCP endpoint — usable with Claude Desktop, Cursor, Windsurf, Cline today.

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`mcp.zylalabs.com/mcp?apikey=YOUR_ZYLA_API_KEY`

7. FULL API ROADMAP — THE CASSANDRA INSTITUTIONAL STACK

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LAYER 1 – Signal Foundation

Cassandra-Core API  LIVE

└ Live RSI, governed registry, Becker-calibrated

└ 54 event families, 3 DOI papers | \$49/mo

LAYER 2 – Intelligence

Cassandra-Pro API  NEXT BUILD (v0.7.0)

└ RSI history (2020–2026), ensemble monetary

└ Theme + family-level decomposition | \$149/mo

LAYER 3 – Fragility Alpha

Cassandra-Alpha API  Paper 4 (Bigdata.com)

└ Fragility Alpha = RSI - VaR delta | \$299/mo

LAYER 4 – Position Sizing

Cassandra-Kelly API  Paper 5

└ Kelly-optimal sizing from governed RSI | \$299/mo

LAYER 5 – Regime Intelligence

Cassandra-Regime API  Paper 6

└ Asset Class RSI Vector, event-driven rotation | \$499/mo

LAYER 6 – On-Chain Cross-Signal

Cassandra-Chain API  Paper 7

└ Stablecoin flows + whale movements × RSI | \$499/mo

LAYER 7 – Bio-Signal (Contagion Model)

Cassandra-Bio API  Paper 8

└ SIR model, $R_0 = \beta/\gamma$, prediction market β | \$999/mo

LAYER 8 – Real-Time Streaming

Cassandra-Stream API  v0.9.0

└ WebSocket live RSI + threshold alerts | \$499/mo

LAYER 9 – Enterprise Suite

Cassandra-Enterprise  v5.0

└ All APIs + dedicated MCP + white-label | Custom

8. THE FRAGILITY ALPHA THESIS (Added 2026-04-02)

Core Insight: VaR is backward-looking. Cassandra RSI is forward-looking. The gap is the alpha.

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Fragility Alpha = RSI_signal - VaR_signal

When Cassandra sees danger VaR doesn't → Advance warning premium

When VaR sees danger Cassandra doesn't → Noise filter

Paper 4 Hypotheses:

- H1: RSI leads VaR spikes by 3–7 trading days on geopolitical events
- H2: RSI-VaR delta predicts abnormal returns better than either signal alone
- H3: Fragility Alpha largest during low-volatility, high-event-hazard regimes

Data requirement: Bigdata.com API (trial confirmed 2026-04-01)

9. ASSET CLASS RSI VECTOR (Layer 5 Detail)

Asset	Kinetic	Sovereign	Trade	Monetary	Technology
SPY	-0.85	-0.70	-0.60	-0.55	-0.40
TLT	+0.60	-0.30	+0.20	-0.80	+0.10
GLD	+0.75	+0.50	+0.30	+0.20	+0.15

BTC	-0.40	-0.60	-0.35	-0.70	-0.80
Cash	+0.50	+0.40	+0.30	+0.25	+0.20

10. ON-CHAIN CROSS-SIGNAL THESIS (Layer 6 Detail)

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On-chain flight + High RSI = CONFIRMED regime stress (highest conviction)

On-chain calm + High RSI = Prediction markets LEADING on-chain (early warning)

On-chain stress + Low RSI = Noise filter – RSI catches false alarm

On-chain calm + Low RSI = Risk-on confirmed (full position)

11. SIR CONTAGION MODEL THESIS (Layer 7 Detail)

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S (Susceptible) = Asset classes exposed to active hazard families

I (Infected) = Assets with RSI below threshold

R (Recovered) = Assets where RSI is normalising post-event

β (Transmission) = Cassandra hazard mass H_t

γ (Recovery) = RSI re-risking speed (~13 days avg)

$R_0 = \beta/\gamma$ = "Contagion Reproduction Number"

$R_0 > 1 \rightarrow$ regime stress reproducing, spreading

$R_0 < 1 \rightarrow$ contagion contained, recovery underway

Novel contribution: No prior framework uses live prediction market probabilities as transmission coefficient β . All prior SIR finance work uses historical price correlations.

12. REVENUE MODEL

Tier	API	Price	Target
1	Cassandra-Core	\$49/mo	Quant developers
2	Cassandra-Pro	\$149/mo	Portfolio managers
3	Cassandra-Alpha	\$299/mo	Risk desks

3	Cassandra-Kelly	\$299/mo	Systematic funds
4	Cassandra-Regime	\$499/mo	Multi-asset allocators
4	Cassandra-Chain	\$499/mo	Crypto-native institutions
4	Cassandra-Stream	\$499/mo	Real-time risk systems
5	Cassandra-Bio	\$999/mo	Institutional risk research
6	Cassandra-Enterprise	Custom	RBC-tier institutions

Conservative Month 6 ceiling: ~\$5,600/mo
Year 2 ceiling (institutional adoption): ~\$25,000/mo

13. MCP STRATEGY

Immediate (free, live): Zyla auto-exposes MCP. Works today with Claude Desktop, Cursor, Windsurf, Cline.

Near-term: Dedicated `cassandra-risk-mcp` Python package on PyPI.

```
bash
pip install cassandra-risk-mcp
json
{
  "mcpServers": {
    "cassandra-risk": {
      "command": "uvx",
      "args": ["cassandra-risk-mcp"],
      "env": {"CASSANDRA_API_KEY": "their-key"}
    }
  }
}
```

14. BIGDATA.COM INTEGRATION PLAN

Status: Trial confirmed 2026-04-01.

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Bigdata.com → Unstructured intelligence (news sentiment, 40,000+ sources)
Cassandra → Structured prediction market RSI
Together → Complete forward-looking risk stack

First experiment: Pull Bigdata.com VaR scores for SPY 2020–2026. Compute delta vs RSI series. Plot lead/lag. This chart is the Tier 3 sales pitch.

15. NEXT BUILD PRIORITIES

text

Immediate:

1. Tier 2 endpoints → v0.7.0
 - GET /v1/rsi/history
 - GET /v1/signals/decomposition
 - GET /v1/ensemble/monetary
 - GET /v1/signals/families
 - GET /v1/risk/overlay?ticker=SPY
 - GET /v1/meta/calibration
2. RapidAPI \$149/mo PRO plan update
3. Zyla listing – add MCP banner

Near-term:

4. cassandra-risk-mcp → PyPI
5. RSI history export from outputs/ backtest series
6. Paper 3 draft – ensemble monetary result

When Bigdata.com trial lands:

7. VaR vs RSI delta computation
 8. Lead/lag analysis (H1: RSI leads VaR by 3–7 days)
 9. Paper 4 draft – Fragility Alpha
-

16. MARKETPLACE EXPANSION ROADMAP

text

- ✅ RapidAPI (live, 8.7 popularity, 1 subscriber)
- ✅ Zyla Labs (live, 100% SLA, MCP-enabled)
- 🕒 APILayer (30 min, free listing)
- 🕒 AWS Marketplace (Tier 2+, Month 2)
- 🕒 Azure Marketplace (Tier 3+, Month 2)

- 🕒 Bigdata.com ecosystem (Enterprise, pending partnership)
 - 🕒 PyPI / MCP Registry
 - 🕒 QuantConnect community
 - 🕒 SSRN author page → API link
 - 🕒 Show HN
-

17. KEY THEORETICAL PROPERTIES

RSI Mathematical Properties (proven):

1. Boundedness: $RSI \in (0, 1]$
2. Strict Monotonicity: rising hazard strictly reduces stability
3. Convex Risk Response: multiple simultaneous risks trigger disproportionate de-risking
4. Interpretability: $Position(t) = RSI(t) \times Position_base$

Soros Bound: Cassandra retains edge below ~15% institutional adoption.

Human Error Beta: -0.80%/yr. Defer to model when $P_human - P_market < 0.30$.

Paranoia Tax: ~1.2% annual drag from false positives. Mathematically justified.

18. REPO STRUCTURE (Key Files)

text

src/cassandra_risk/

- becker_calibration.py
- monetary_subablation.py
- geopolitical_subbucket.py
- promotion_workflow.py
- signal_contract.py

data/governed/

- signal_registry.json
- promotion_audit.csv

outputs/

- becker_stack_summary.csv
- risk_decomposition.csv
- monte_carlo/

docs/adr/

- ADR-005-calibration-architecture.md

19. SESSION STATE (as of 2026-04-02, 00:05 IST)

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Last clean tag: v0.5.7-geo-subbucket-calibration (8ec3f50)

Live API: v1.0.0 on Railway

Marketplace: RapidAPI  | Zyla 

Subscribers: 1 (RapidAPI Basic)

Bigdata.com: Trial confirmed

MCP: Zyla auto-MCP live | Dedicated MCP = next build

Active papers:

- Paper 1  DOI live
- Paper 2  DOI live
- Paper 3  In progress
- Paper 4  Pending Bigdata.com trial

Next tag: v0.7.0-tier2-endpoints

Next session goal: Tier 2 API build

20. CASSANDRA × QCA INTEGRATION THESIS (Added 2026-04-02)

The Structural Resonance

Both frameworks use structured ex-ante observables to detect fragility before it becomes visible in price space.

Property	Cassandra RSI	QCA QII
Domain	Macro regime fragility	Firm-level event alpha
Scale	Market-wide	Single ticker
Output scalar	$RSI \in (0,1]$	$QII \in [-1,1]$
Fragility measure	Hazard mass H_t	$d_{eff} = 2^{\wedge}S$
Information set	Forward-looking (prediction markets)	Strictly ex-ante (t-1 close)
Published papers	2 DOI-live, 1 in progress	9 DOI-live

The Blind Spot Each Has Alone

- **Cassandra without QCA:** Knows macro is fragile. Cannot tell which firm-level events to trade within that regime.
- **QCA without Cassandra:** Knows announcement quality and coherence time. Doesn't know if macro will destroy that coherence window before the signal can run.

Four Mathematical Integration Points

Connection 1 — RSI as Superior Phase Input

QCA phase angle θ already includes Macro_{t-1} (VIX + rate vol + USD). Cassandra RSI is strictly superior because it is forward-looking while VIX is backward-looking.

$$\theta_{\text{upgraded}} = \theta_{\text{base}} \times (1 - \alpha \times (1 - \text{RSI}_t))$$

Testable hypothesis: replacing VIX with RSI in θ construction improves QII predictive R^2 .

Connection 2 — Regime-Adjusted Coherence Time

Base coherence law: $\tau = 180 \times e^{(-0.035\theta)}$, fitted at $R^2 = 0.84$.

$$\tau_{\text{adjusted}} = \tau_{\text{QII}} \times \text{RSI}_t$$

Example: Apple, $\theta = 68^\circ$, base $\tau = 22$ days, $\text{RSI} = 0.55 \rightarrow \tau_{\text{adjusted}} = \mathbf{12 \text{ days.12 days.}}$
Shorten holding horizon, not position conviction.

Connection 3 — Unified Fragility Index

$$F_{\text{unified}} = (1 - \text{RSI}_t) \times (d_{\text{eff}} - 1) / 4$$

Oracle Nov 2025: $\text{RSI} = 0.45$, $d_{\text{eff}} = 4.31 \rightarrow F_{\text{unified}} = 0.55 \times 0.83 = \mathbf{0.46}$ (red zone).
Drawdown that followed: 47.7%.

Connection 4 — Multi-Scale Contagion R_0

$$R_{0\text{unified}} = \beta_{\text{macro}} \times \bar{\rho}_{ij} \times \bar{d}_{\text{eff_normalized}}$$

- β_{macro} = Cassandra H_t (macro transmission, from prediction markets)
- $\bar{\rho}_{ij}$ = mean QCA cross-firm coupling (sector transmission)
- $\bar{d}_{\text{eff_normalized}}$ = mean firm fragility across entangled cluster

Three-scale contagion: **macro** $\rightarrow$ **sector** $\rightarrow$ **firm**. When $R_{0\text{unified}} > 1$, regime stress reproduces across all scales simultaneously.

Unified API Response

json

GET /v1/unified/prescreen?ticker=NVDA&event_date=2026-05-15

```
{
  "macro_layer": {
    "cassandra_rsi": 0.71, "hazard_mass": 0.41,
    "regime": "moderate_fragility", "dominant_hazard": "monetary"
  },
  "firm_layer": {
    "qii_score": 0.58, "theta_degrees": 52,
    "archetype": "wave_dominant", "deff": 3.46
  },
  "unified_layer": {
    "fragility_index": 0.21,
    "base_coherence_days": 22,
    "regime_adjusted_coherence_days": 15.6,
    "position_scalar": 0.57,
    "key_risk": "macro_shortening_coherence_window"
  },
  "contagion_layer": {
    "bio_r0_unified": 1.18,
    "cascade_risk": "elevated"
  }
}
```

The Bridge Paper

Paper CxQ: "Cassandra × QCA: Regime-Conditioned Event Alpha, Unified Fragility Index, and Multi-Scale Contagion"

- Cross-listed in both series simultaneously
- Permanent bridge paper — cited by all subsequent papers in both series
- Target: Journal of Risk, Risk Magazine, JFQA

The Quantumist Intelligence Stack

text

Layer 1 (Macro):	Cassandra RSI	← regime fragility
Layer 2 (Event):	QII Prescreen	← event quality
Layer 3 (Time):	Regime-Adjusted τ	← when to exit
Layer 4 (Network):	QCA Contagion p_{ij}	← sector exposure
Layer 5 (Unified):	$F_{unified}$	← joint fragility
Layer 6 (Bio):	$R_0_{unified}$	← cross-scale spread
Layer 7 (Position):	CxQ Position Scalar	← $RSI \times (QII+1)/2$

21. QCA API DEPLOYMENT PLAN (quantumist.ai/api)

Why QCA Deploys Faster Than Deep Cassandra Tiers

- 9 published papers = zero research debt
- All inputs computable from public sources: SEC filings (P), Yahoo Finance options (W), pre-event NLP (θ)
- No live prediction market ingestion needed
- Same Railway + Zyla + RapidAPI playbook already proven

Phase 1 Endpoints (Deploy in 1 Session)

text

GET /v1/qii/prescreen?ticker=AAPL&event_date=2026-05-01

GET /v1/qii/archetype?ticker=NVDA

GET /v1/coherence/tau?ticker=AMZN&theta=42

GET /v1/fragility/deff?ticker=ORCL

Phase 2 Endpoints (Week 2)

text

GET /v1/contagion/network ← full 7×7 p_{ij} matrix

GET /v1/unified/prescreen ← all 4 layers, the flagship

Pricing (QCA Standalone)

Tier	Endpoints	Price
QII-Basic	Prescreen + archetype	\$49/mo
QII-Pro	+ Coherence τ + d_{eff}	\$149/mo
QII-Contagion	+ p_{ij} network	\$299/mo
QII-Enterprise	Full suite + unified	Custom

Pricing (Unified Cassandra × QCA)

Tier	Description	Price
Unified-Pro	Cassandra Pro + QII-Pro	\$249/mo
Unified-Regime	Full CxQ stack, all 7 layers	\$599/mo

arXiv Abstract Line (Copy-Paste for All 9 QCA Papers)

"All results independently verifiable via the Quantumist Intelligence API (api.quantumist.ai, RapidAPI: quantumist-intelligence). Readers may compute live QII scores, archetype classifications, coherence windows, and entanglement risk screens without any local installation — including via MCP-native natural language access."

Codex Instruction (QCA API Build)

text

1. New Railway service: quantumist-qca-api
2. FastAPI scaffold (mirror cassandra-risk structure)
3. Implement:
 - $QII = (2 * P * W / (P^2 + W^2)) * \cos(\theta \times \pi / 180)$
 - $\tau = 180 * \exp(-0.035 * \theta)$
 - $deff = 2^S$ (eigenvalue entropy of 5x5 pre-event correlation matrix)
 - archetype classifier (7 boundary conditions, Paper IX §2.8)
4. Deploy → qca-api.railway.app
5. List: RapidAPI (quantumist-intelligence) + Zyla (auto-MCP)
6. Tag: v1.0.0-qca-core
7. Add API link to all 9 QCA paper abstracts on ResearchGate
8. Submit Papers I–IX to arXiv q-fin.PM + q-fin.RM

Session State for QCA Build

text

Research: ☒ Complete (9 papers, Paper IX axiomatic)
Math: ☒ All formulae in Paper IX appendix
Playbook: ☒ Railway + Zyla + RapidAPI proven via Cassandra
Next tag: v1.0.0-qca-core
Timeline: Phase 1 live in 1 session

v2.0 — Updated 2026-04-02 01:00 IST

Maintained by Umran Nayani.

Commit to: [docs/CASSANDRA_MASTER_CONTEXT_v2.md](#)
